

CAMPBELL AAF, KY, USA

WMO: 746710

Lat: 36.667N

Lon: 87.483W

Elev: 573

StdP: 14.39

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13806

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.1	16.5	1.3	6.0	15.3	6.9	8.0	19.9	24.3	50.6	21.4	45.4	6.6	350	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	94.0	76.0	91.5	75.7	90.0	75.2	79.4	89.1	78.3	87.5	77.4	86.1	8.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.0	143.8	83.3	75.6	136.9	82.4	74.9	133.7	81.8	43.2	89.2	42.2	87.7	41.1	85.8	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	18.1	16.0	4.0	97.0	7.3	3.9	-1.2	99.8	-5.5	102.1	-9.6	104.3	-14.9	107.1
			3.9	81.8	7.7	1.5	-1.6	82.9	-6.1	83.7	-10.4	84.6	-15.9	85.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.2	36.6	41.4	49.5	59.4	68.4	75.9	78.8	77.9	71.5	60.4	48.6	40.6	(6)
(7)		DBStd	17.23	12.37	11.76	10.80	9.17	7.30	5.10	4.36	4.55	6.78	8.56	9.56	10.58	(7)
(8)		HDD50	1378	445	287	144	24	0	0	0	0	13	138	327		(8)
(9)		HDD65	3769	882	663	490	212	55	2	0	21	193	495	756		(9)
(10)		CDD50	4728	28	46	128	307	569	778	892	866	646	336	96	36	(10)
(11)		CDD65	1648	0	1	10	45	159	330	427	402	217	52	4	1	(11)
(12)		CDH74	15755	0	5	91	389	1314	3134	4324	3975	2020	477	25	1	(12)
(13)		CDH80	6210	0	0	12	64	369	1247	1880	1737	788	111	2	0	(13)
(14)	Wind	WSAvg	6.8	8.1	8.2	8.1	8.2	6.8	6.1	5.5	5.2	5.3	6.1	6.9	7.5	(14)
(15)	Precipitation	PrecAvg	48.8	4.0	3.9	4.5	4.4	5.2	4.0	4.3	2.9	3.3	3.7	3.9	4.8	(15)
(16)		PrecMax	58.3	9.4	8.5	13.0	8.0	11.5	11.9	12.0	9.6	9.7	8.9	9.3	12.6	(16)
(17)		PrecMin	36.9	1.4	1.4	1.9	1.7	1.6	0.8	1.0	0.3	0.5	0.0	0.8	1.4	(17)
(18)		PrecStd	6.2	1.7	1.9	2.3	1.9	2.2	2.2	2.4	1.9	2.2	2.2	2.2	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.5	73.2	81.5	84.8	90.3	94.3	97.4	98.9	95.1	87.8	77.3	71.6	(19)
(20)		0.4%	MCWB	62.2	61.3	63.5	67.4	73.1	76.7	77.7	75.7	72.1	70.8	64.1	63.9	(20)
(21)		2%	DB	65.4	68.8	76.2	81.8	86.4	91.1	93.4	94.2	91.1	82.9	72.5	65.8	(21)
(22)		2%	MCWB	60.7	60.0	61.5	65.9	71.8	75.8	77.5	75.6	72.5	68.4	61.0	60.5	(22)
(23)		5%	DB	61.1	64.4	72.0	78.8	84.1	89.5	91.2	91.1	88.3	79.4	68.3	61.5	(23)
(24)		5%	MCWB	56.4	56.4	59.1	64.1	70.5	75.3	77.1	75.4	72.1	65.7	59.7	56.7	(24)
(25)		10%	DB	56.2	60.3	67.4	75.2	81.7	87.3	89.4	88.9	84.7	75.4	64.1	57.3	(25)
(26)		10%	MCWB	50.6	53.4	56.9	62.7	69.7	74.5	76.5	74.7	70.8	64.4	57.4	52.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.5	65.1	66.5	71.4	76.5	80.0	81.3	80.5	77.5	74.1	67.5	65.4	(27)
(28)		0.4%	MCDWB	66.8	68.9	75.7	80.5	85.0	89.7	91.0	90.6	86.4	81.4	72.3	69.0	(28)
(29)		2%	WB	61.4	62.4	64.0	69.1	74.4	78.3	79.8	78.9	75.8	71.8	64.6	61.8	(29)
(30)		2%	MCDWB	64.1	67.4	72.6	77.5	83.0	87.8	89.5	88.3	84.2	79.0	69.4	65.0	(30)
(31)		5%	WB	57.6	58.7	61.6	67.1	72.8	77.2	78.7	77.8	74.4	69.4	61.8	58.0	(31)
(32)		5%	MCDWB	61.2	62.9	68.8	74.8	80.9	86.2	88.0	86.6	82.8	75.2	66.6	60.8	(32)
(33)		10%	WB	51.8	54.4	58.9	64.9	71.3	75.9	77.7	76.7	73.1	67.0	58.5	53.5	(33)
(34)		10%	MCDWB	55.2	59.2	66.2	72.1	78.6	84.4	86.6	84.8	81.2	73.4	63.4	56.8	(34)
(35)	Mean Daily Temperature Range	MDBR	16.9	18.4	20.4	21.5	19.1	18.7	18.5	19.8	21.6	21.9	19.7	16.4	(35)	
(36)		5% DB	MCDBR	20.4	23.6	25.7	23.9	21.3	20.7	20.8	22.7	24.8	25.4	24.3	20.6	(36)
(37)		5% DB	MCWBR	16.0	16.3	14.5	12.4	9.6	8.5	8.0	8.3	9.1	12.4	15.3	15.8	(37)
(38)		5% WB	MCDBR	18.5	20.0	22.2	19.6	18.2	18.3	18.4	18.6	18.9	19.9	19.7	17.9	(38)
(39)		5% WB	MCWBR	16.1	16.6	14.8	12.0	9.3	8.6	8.2	8.1	8.2	11.5	15.1	16.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.323	0.371	0.400	0.439	0.458	0.495	0.496	0.421	0.366	0.329	0.309	(40)	
(41)		taud	2.480	2.439	2.348	2.269	2.208	2.180	2.098	2.083	2.284	2.426	2.481	2.505	(41)	
(42)		Ebn at Noon	279	290	283	280	269	263	252	249	263	271	271	273	(42)	
(43)		Edh at Noon	26	31	37	42	46	47	50	50	39	31	26	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	687	874	1216	1588	1805	2007	1923	1765	1485	1135	801	584	(44)	
(45)		RadStd	61	112	99	102	155	136	112	84	105	127	83	56	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+128	(47)

Nomenclature: See separate page